

## Code-Layout, Readability and Reusability

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|----------------------|---------------|
| <b>Date</b>          | 30 June 2026  |
| <b>Team ID</b>       | XXXXXX        |
| <b>Project Name</b>  | RISING WATERS |
| <b>Maximum Marks</b> | 5 Marks       |

### Code-Layout, Readability and Reusability:

This document evaluates the Rising Waters project in terms of code structure, readability, maintainability, and reusability. The project follows proper coding standards, meaningful naming conventions, modular design, and documentation to ensure that the system is easy to understand, maintain, and extend for future development.

#### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks                        |
|------|---------------------------|-----------------------------------------------|-------------------------------|--------------------------------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes                           | Proper indentation followed    |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes                           | Files organized into folders   |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           | Meaningful variable names used |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           | Functions properly named       |
| 5    | Code Comments             | Inline and block comments explain logic       | Yes                           | Comments added where required  |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes                           | Modular code design followed   |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes                           | No redundant code              |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes                           | Error handling implemented     |

### Reusable Components / Modules:

| S.No | Component / Module Name | Language / Technolog            | yWhere Reused            | Reusability Level (High / Medium / Low) |
|------|-------------------------|---------------------------------|--------------------------|-----------------------------------------|
| 1    | User Authentication     | Python, Flask                   | Login & Registration     | High                                    |
| 2    | Water Level Monitoring  | Python                          | Dashboard                | High                                    |
| 3    | Flood Prediction Model  | Machine learning (scikit-learn) | Prediction Module        | High                                    |
| 4    | Alert Notification      | Python                          | SMS / Email Alerts       | Medium                                  |
| 5    | Database Module         | MySQL                           | Uder & Flood Data Storge | High                                    |

### Overall Code Quality Assessment:

| Aspect                   | Rating (1-5) | Comments                     |
|--------------------------|--------------|------------------------------|
| Code Layout & Structure  | 5            | Well Organized               |
| Readability              | 5            | Easy to Understand           |
| Reusability              | 5            | Modules can be Reused        |
| Documentation / Comments | 4            | Sufficient comments provided |
| Overall Score            | 5/5          | Good code quality            |